Cardiovascular System (CARD) · NPLEX-P1-CARD-12

CV-12 Exercise and the Cardiovascular System

Work each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Space the two brain dumps for one competency across different days rather than doing them back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-12-01-C01				
NPLEX-P1-CARD-12-01-C02				
NPLEX-P1-CARD-12-01-C03				
NPLEX-P1-CARD-12-01-C04				
NPLEX-P1-CARD-12-01-C05				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-12-01-C01-BD1

Draw the two inputs that start the exercise response

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-12 Exercise and the Cardiovascular System
Competency: NPLEX-P1-CARD-12-01-C01 Activity: NPLEX-P1-CARD-12-01-C01-BD1 Date: _____

Without looking at your notes, draw the contracting muscle, the medullary cardiovascular centers, and the heart, and show the two signals that raise heart rate at the onset of exercise. Label the feedforward signal and write where it comes from, what its output scales with, and what it does to the arterial baroreflex set point. Label the feedback reflex, then split it into its two arms: for each arm write the afferent fiber class, the stimulus that excites it, when during the contraction it is strongest, and its main cardiovascular effect. Finish by writing the two experimental observations from the notes that show central command and the metaboreflex are separate drivers.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-12-01-C01-BD2

Why the rate rise is vagal first and sympathetic second

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-12 Exercise and the Cardiovascular System
Competency: NPLEX-P1-CARD-12-01-C01 Activity: NPLEX-P1-CARD-12-01-C01-BD2 Date: _____

Write the receptor, the G protein, the second messenger or subunit, and the channel for vagal slowing and for sympathetic acceleration of the SA node, then explain from how each transmitter is cleared why heart rate rises within one or two beats when vagal firing stops but sympathetic effects take seconds. Give the approximate heart rate up to which vagal withdrawal accounts for most of the rise, and write what sympathetic drive adds above that rate. Then write the three mechanisms that raise venous return so cardiac output can actually rise, and name the relationship that converts greater filling into a larger stroke volume. Finish with three people starting the same exercise: one has a denervated transplanted heart, one is taking a beta-1 blocker, and one has heart failure. For each, write what happens to heart rate at the onset, how cardiac output rises instead, and why.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-12-01-C01-RT01

Rebuild the onset sequence

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-12 Exercise and the Cardiovascular System
Competency: NPLEX-P1-CARD-12-01-C01 Activity: NPLEX-P1-CARD-12-01-C01-RT01 Date: _____

Rebuild the seven-step sequence that runs from the recruitment of motor units to a matched rise in muscle blood flow, one step per line and in order. Beside the step where heart rate first rises, write the channel that closes and how long the change takes. Beside the sympathetic step, write the three things sympathetic outflow is doing at once. Beside the venous return step, write the two pumps involved and what each one does mechanically. Beside the final step, write which afferent fibers are responsible and what condition in the muscle brings them in.

Rebuild from memory: Cardiovascular response at the onset of exercise

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-12-01-C02-BD1

Write the Fick principle and fill in the numbers

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-12 Exercise and the Cardiovascular System
Competency: NPLEX-P1-CARD-12-01-C02 Activity: NPLEX-P1-CARD-12-01-C02-BD1 Date: _____

Write the Fick principle as an equation for the whole body and define each term. Write the equation for arterial O2 content and work it out for a hemoglobin of 15 g/dL at a saturation of 98%. Then build a table from memory with rows for heart rate, stroke volume, cardiac output, mixed venous O2 content, the arteriovenous O2 difference, and VO2, and columns for rest, maximal exercise in an untrained young adult, and maximal exercise in an elite endurance athlete. Fill every cell with the approximate value from the notes. Underneath, write the three reasons O2 extraction rises during exercise, and name the effect that shifts the oxyhemoglobin curve in active muscle.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-12-01-C02-BD2

Work VO2max from the numbers, then take the hemoglobin away

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-12 Exercise and the Cardiovascular System
Competency: NPLEX-P1-CARD-12-01-C02 Activity: NPLEX-P1-CARD-12-01-C02-BD2 Date: _____

A 25-year-old untrained man weighing 75 kg has a resting heart rate of 70/min, a stroke volume of 70 mL, an arterial O2 content of 20 mL/dL, and a mixed venous O2 content of 15 mL/dL. At maximal effort his heart rate is 195/min, his stroke volume is 110 mL, and his mixed venous content falls to 5 mL/dL. Calculate resting VO2 and VO2max in mL/min and in mL/kg/min, showing your working, and state by what factor flow rose and by what factor extraction rose. Now repeat the calculation with a hemoglobin of 9 g/dL and a mixed venous content of 4 mL/dL at maximum, and write the cardiac output that would be needed to hold the original VO2max. Explain from your two sets of numbers what happens to VO2max and to heart rate at a fixed submaximal workload, and name the variable that limits VO2max in a healthy person at sea level. Finish by writing why stroke volume plateaus at about 40% to 60% of VO2max in an untrained person and what carries cardiac output higher after that point.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-12-01-C03-BD1

Rebuild the distribution of cardiac output

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-12 Exercise and the Cardiovascular System
Competency: NPLEX-P1-CARD-12-01-C03 Activity: NPLEX-P1-CARD-12-01-C03-BD1 Date: _____

From memory, build a table of where cardiac output goes at rest and at maximal exercise, with one row for skeletal muscle, coronary, brain, skin, splanchnic, and kidney. Use columns for resting flow with its percentage of a total of about 5.8 L/min, maximal exercise flow against a total of about 25 L/min, and the control that dominates that bed. Fill every cell. Underneath, write the resting and peak flow per 100 g of skeletal muscle and the fold increase. Then list every local mediator the notes give for dilation in active muscle and write in one line what each one does, including which channel interstitial K+ acts through and what makes red cells release ATP.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-12-01-C03-BD2

Why total resistance falls while some beds constrict

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-12 Exercise and the Cardiovascular System
Competency: NPLEX-P1-CARD-12-01-C03 Activity: NPLEX-P1-CARD-12-01-C03-BD2 Date: _____

Sympathetic outflow rises to every vascular bed during exercise, yet flow to active muscle increases up to 25 fold. Explain how both are true, naming the process that blunts alpha-adrenergic constriction in contracting muscle, the receptor subtype most involved, and why some residual constriction has to remain. Then explain from the arrangement of the vascular beds why total peripheral resistance falls even though gut, kidney, and inactive muscle arterioles are constricting. Next write what happens to coronary flow and why extraction cannot supply the extra demand, then write what that predicts for a person with a coronary stenosis that causes no symptoms at rest. Finish by writing what happens to renal blood flow, to GFR, and therefore to filtration fraction, and why cerebral flow is the exception.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-12-01-C03-RT01

Rebuild how flow concentrates in working muscle

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-12 Exercise and the Cardiovascular System
Competency: NPLEX-P1-CARD-12-01-C03 Activity: NPLEX-P1-CARD-12-01-C03-RT01 Date: _____

Rebuild the six-step sequence that explains how flow is concentrated in working muscle, one step per line and in order. Beside the step on inactive beds, name the receptor that mediates the constriction and list the beds involved. Beside the step on active muscle, list the local mediators. Beside the step on sympatholysis, write which signals do the blunting. Beside the final step, write the one variable all the parallel beds share and the equation that converts a bed's resistance into its share of flow.

Rebuild from memory: How flow concentrates in working muscle

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-12-01-C03-PR01

Predict what heat does to a steady run

A healthy adult runs at the same speed for 60 minutes on a cool day and then on a hot, humid day. Core temperature on the hot day rises to 39 °C (102.2 °F). For each item below, write the direction of the change on the hot day and give the mechanism in one line.

What changes	Up, down, or no change	Why, in one or two lines
What happens to skin blood flow at a heavy submaximal workload, and which nerve fibers produce it		
What happens to the volume of blood available to fill the heart		
What happens to stroke volume over the 60 minutes		
What happens to heart rate over the 60 minutes, and what this pattern is called		
What happens to the flow available to the leg muscles		
What happens to skin blood flow if the runner pushes on to maximal effort, and what that costs		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-12-01-C04-BD1

Pressure in dynamic exercise against pressure in static exercise

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-12 Exercise and the Cardiovascular System
Competency: NPLEX-P1-CARD-12-01-C04 Activity: NPLEX-P1-CARD-12-01-C04-BD1 Date: _____

Write the equation that relates mean arterial pressure to cardiac output and total peripheral resistance. Then, for dynamic exercise and for static exercise in turn, write what happens to heart rate, stroke volume, cardiac output, total peripheral resistance, systolic pressure, diastolic pressure, and pulse pressure, and name the main pressor drive in each. Give the usual peak systolic pressure range in healthy adults during dynamic exercise. For static exercise, explain what rising intramuscular pressure does to the muscle's own blood vessels and how that produces the pressor response. Finish by naming the type of load each form of exercise puts on the left ventricle and writing what happens to the shape of the pressure-volume loop in each.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-12-01-C04-BD2

Read pressures and work out resistance

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-12 Exercise and the Cardiovascular System
Competency: NPLEX-P1-CARD-12-01-C04 Activity: NPLEX-P1-CARD-12-01-C04-BD2 Date: _____

Write the formula for mean arterial pressure from systolic and diastolic pressure, and the formula for total peripheral resistance from mean arterial pressure and cardiac output. Then calculate mean arterial pressure and total peripheral resistance in mmHg per L/min for three states, showing your working: at rest with 120/80 mmHg and a cardiac output of 5 L/min; while running with 180/76 mmHg and a cardiac output of 20 L/min; and during a sustained handgrip with 165/110 mmHg and a cardiac output of 6.5 L/min. Write one sentence for each state explaining why pressure moved the way it did, and state which calculation is least accurate at high heart rates and why. Then write the index of myocardial O2 demand given in the notes and use it to explain why a static effort can provoke ischemia at a lower heart rate than running. Finish by writing what a fall in systolic pressure during a rising dynamic workload means and naming three conditions that produce it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-12-01-C04-CP01

Dynamic against static exercise

Fill in every cell from memory, then check it against the notes and correct what you missed.

Feature	Dynamic exercise such as running or cycling	Static exercise such as a handgrip or heavy hold
Heart rate		
Stroke volume		
Cardiac output and how it rises		
Total peripheral resistance		
Systolic pressure		
Diastolic pressure and pulse pressure		
Main pressor drive		
Load on the left ventricle and shape of the pressure-volume loop		
Effect of a sustained handgrip on the murmurs of mitral regurgitation and of hypertrophic cardiomyopathy		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-12-01-C05-BD1

Eccentric and concentric remodeling, and the trained heart

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-12 Exercise and the Cardiovascular System
Competency: NPLEX-P1-CARD-12-01-C05 Activity: NPLEX-P1-CARD-12-01-C05-BD1 Date: _____

Write how adult myocytes respond to a chronic load and why. Define eccentric and concentric hypertrophy by where the sarcomeres are added, what happens to cavity size and wall thickness, and which kind of load produces each, then name the relationship that explains why both bring wall stress back toward normal. Next describe the endurance-trained left ventricle: wall thickness, wall-to-cavity ratio, right ventricle and atria, and diastolic filling. Write what happens to plasma volume and red cell mass with training and why hemoglobin concentration can fall. Then explain, using a resting cardiac output of about 5 L/min and a stroke volume of about 100 mL, why the resting heart rate is about 50/min, name the main mechanism of the slowing, and list the three benign conduction findings that come from the same mechanism and what happens to them with exercise. Finish by writing what training changes and what it does not change at a fixed submaximal workload and at maximal effort.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-12-01-C05-BD2

Separate physiologic growth from sarcomere disease

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-12 Exercise and the Cardiovascular System
Competency: NPLEX-P1-CARD-12-01-C05 Activity: NPLEX-P1-CARD-12-01-C05-BD2 Date: _____

Write the growth factor and the intracellular pathway that drive physiologic hypertrophy, and the neurohormones, receptor class, and signaling pathway that drive pathologic hypertrophy, then write what pathologic growth does to the fetal gene program, the capillary supply, the interstitium, and ventricular function. Next write the inheritance pattern of hypertrophic cardiomyopathy and the two genes most often involved, then explain the mechanism behind each of these three features: exertional dyspnea, angina with normal epicardial coronary arteries, and sudden death in a young athlete. Then write what causes dynamic outflow obstruction and list the four changes in loading or contractility that worsen it, with the bedside maneuvers that make the murmur louder and softer and why. Finish by writing what happens to each heart after about three months without training and why that result separates the two conditions.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-12-01-C05-CP01

Athlete's heart against hypertrophic cardiomyopathy

Fill in every cell from memory, then check it against the notes and correct what you missed.

Feature	Endurance athlete's heart	Hypertrophic cardiomyopathy
Wall thickness and whether it is symmetric		
Cavity dimension		
Diastolic function		
Histology		
Outflow tract		
Electrocardiogram		
Peak VO2		
Result after about three months of detraining		
Family history		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-12-01-C05-CA01

Near syncope in a young athlete during a hot race

An 18-year-old cross-country runner is seen after nearly fainting in the final kilometer of a race run at 32 °C (89.6 °F). He had a tight sensation in the chest for the last several minutes and felt his heart pounding. He has run competitively for four years and trains daily. A maternal uncle died suddenly at 31 years of age. Temperature is 38.6 °C (101.5 °F), pulse is 58/min and regular, respirations are 16/min, and blood pressure is 112/74 mm Hg. A systolic murmur is heard at the left sternal border; it becomes louder when he stands up from squatting and softer during a sustained handgrip. The electrocardiogram shows sinus bradycardia with deep T wave inversions in leads V4 through V6 and Q waves in leads I and aVL. Echocardiography shows a septal wall thickness of 18 mm, a posterior wall thickness of 10 mm, and a left ventricular cavity dimension of 42 mm, with impaired relaxation on diastolic filling.

1. Recognize. Name the structural diagnosis and the type of protein the underlying mutations affect, and write its usual inheritance pattern.

2. Explain. Explain from the stiff ventricle and the muscle mass why he had chest tightness and breathlessness during the race, and why the epicardial coronary arteries can be normal.

3. Differentiate. Write the features in this case that separate it from the heart of a trained endurance runner, taking wall thickness, symmetry, cavity size, diastolic filling, electrocardiogram, and family history in turn.

4. Investigate. Write which measurements and observations you would use to confirm the diagnosis, and what value or appearance each one would have in this condition rather than in a trained heart.

5. Decide. State which physiologic change during a hot race made the outflow gradient worse, and which mechanism accounts for the risk of sudden death.

6. Justify. Justify your answer from the effect of preload, afterload, contractility, and heart rate on ventricular size, and explain why the murmur got louder on standing and softer during handgrip.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-12-01-C02-Q01

Accounting for the rise in oxygen uptake

A 24-year-old untrained man undergoes a graded cycling test. At rest his cardiac output is 5 L/min, his arterial O2 content is 20 mL/dL, and his mixed venous O2 content is 15 mL/dL. At peak effort his cardiac output is 21 L/min, his arterial O2 content is 20 mL/dL, and his mixed venous O2 content is 5 mL/dL. Which of the following best accounts for the increase in O2 uptake from rest to peak effort?

A. A rise in arterial O2 content produced by more efficient pulmonary gas exchange

B. An approximately fourfold rise in cardiac output together with an approximately threefold rise in O2 extraction

C. A rise in cardiac output with no change in the arteriovenous O2 difference

D. A widening of the arteriovenous O2 difference while cardiac output stays at its resting value

Why did you pick it?

Board-style question NPLEX-P1-CARD-12-01-C03-Q02

Regional flow during submaximal cycling

A healthy 30-year-old woman cycles at 70% of her maximal O2 uptake in a thermoneutral room. Which of the following changes in regional blood flow is expected during the exercise?

A. Coronary blood flow rises about three- to fourfold, because myocardial O2 extraction is already near maximal at rest

B. Renal blood flow rises in proportion to the rise in cardiac output

C. Cerebral blood flow rises to about three times its resting value

D. Splanchnic blood flow is unchanged, because local metabolic control overrides sympathetic constriction in the gut

Why did you pick it?

Board-style question NPLEX-P1-CARD-12-01-C04-Q03

Hemodynamics of a sustained handgrip

A healthy 28-year-old man holds a handgrip at 40% of his maximal voluntary force for two minutes. Which of the following sets of changes is expected at the end of the two minutes, compared with his resting values?

A. Total peripheral resistance falls, diastolic pressure falls slightly, and stroke volume rises markedly

B. Cardiac output rises about fourfold and pulse pressure widens while diastolic pressure is unchanged

C. Heart rate falls as the arterial baroreflex buffers the rise in pressure, and cardiac output is unchanged

D. Total peripheral resistance is unchanged or higher, and systolic and diastolic pressure both rise sharply

Why did you pick it?

Board-style question NPLEX-P1-CARD-12-01-C05-Q04

Reading an echocardiogram in a trained cyclist

A 19-year-old competitive cyclist is evaluated after a resting pulse of 46/min is found at a preparticipation examination. He has no symptoms and no family history of sudden death. The electrocardiogram shows sinus bradycardia with isolated high QRS voltage and upright T waves. Echocardiography shows a symmetric left ventricular wall thickness of 11 mm, a left ventricular cavity dimension of 58 mm, an enlarged right ventricle, and enhanced early diastolic filling. Which of the following findings most strongly supports physiologic adaptation to endurance training rather than hypertrophic cardiomyopathy?

A. The resting heart rate of 46/min

B. The isolated high QRS voltage on the electrocardiogram

C. The enlarged cavity with symmetric wall thickening and enhanced diastolic filling

D. The symmetric wall thickness of 11 mm taken on its own

Why did you pick it?